

One acid + one amine, two reactions the library will sell you

Templates 136–139 (“direct acid → amine”, yield 0.70) leave the carbonyl oxygen UNMAPPED because it leaves: the product is a secondary amine.

Templates 76–79 (“Amide synthesis”, yield 0.75) map it through and give the amide. Identical reactants, so the model chooses between them — and the O₂ loss with hydrogens balanced is the signature of the reduction, not of a bug.

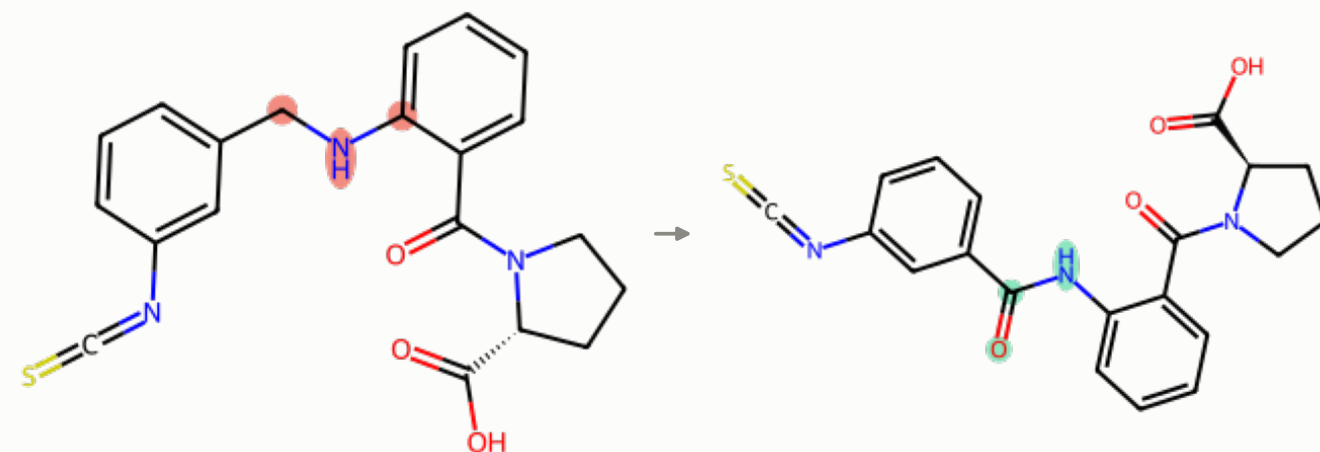
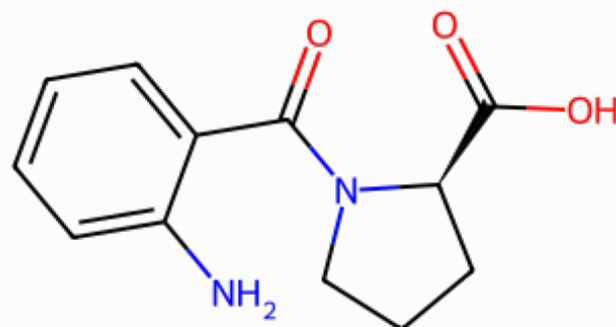
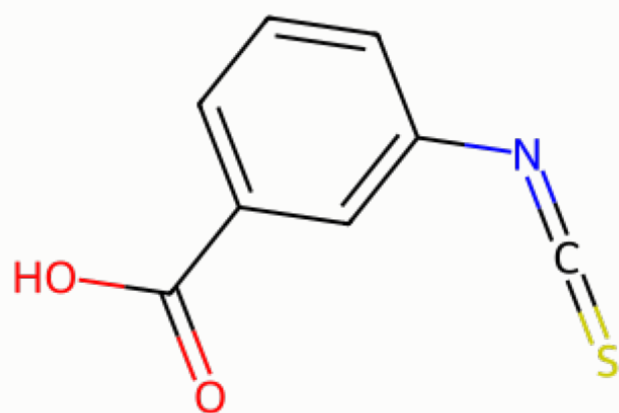
acid reactant

amine (the growing molecule)

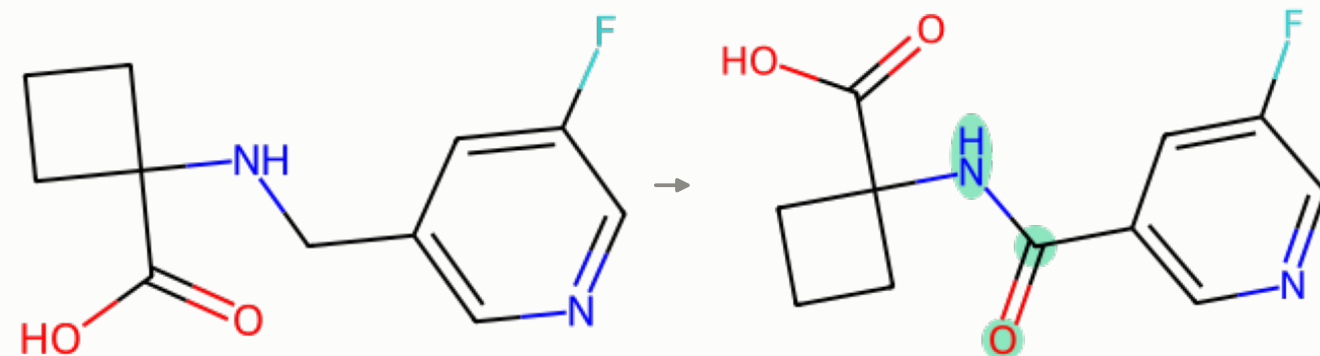
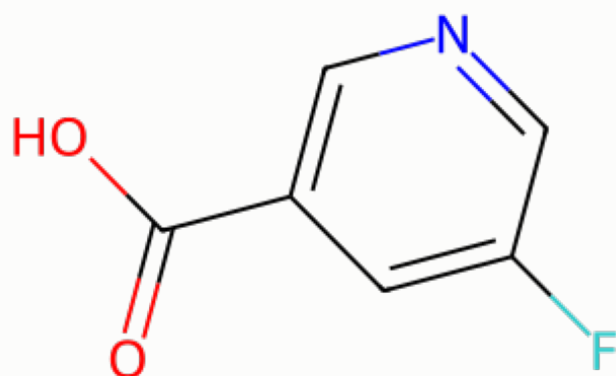
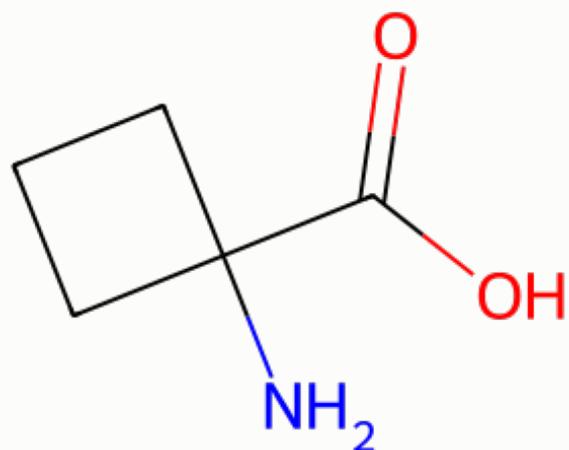
direct acid → amine (0.70) — recorded

amide coupling (0.75) — same reactants

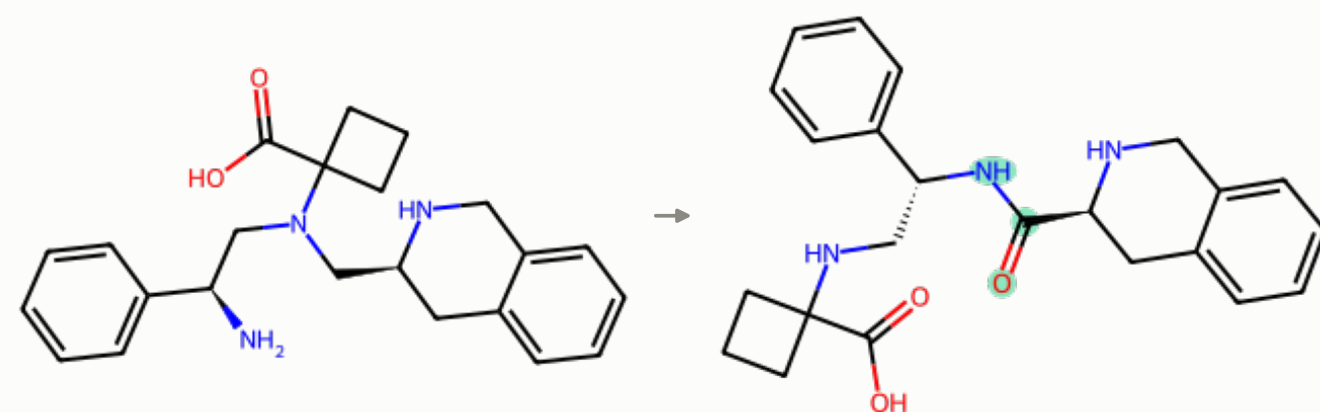
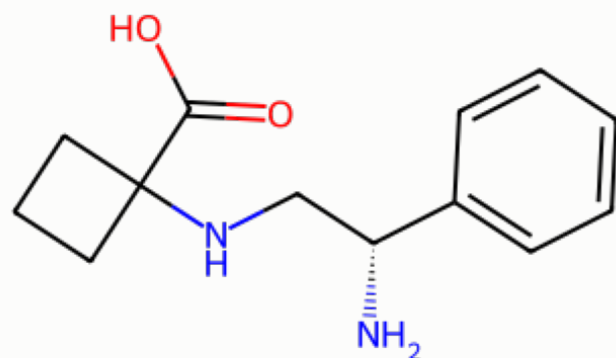
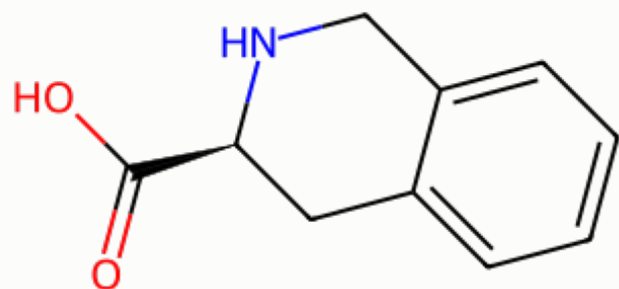
template 137
“direct acid → amine”



template 136
“direct acid → amine”



template 139
“direct acid → amine”



Both routes are valid synthesis instructions; they are not equally easy to run, and the reaction count we budget against cannot tell them apart. The direct reduction is 267,204 of 747,472 steps (35.7%) across the four publishable cells — 99.8% of DRD2’s acid+amine steps, but only 31.7–42.3% of sEH’s. Logs/071 measures why.